CSE 4213: Pattern Recognition

Lecture 8 and 9
(Principal Components Analysis)

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Principal Components Analysis (PCA)

→ PCA is a useful statistical technique for finding patterns in data of high dimension.

→ It is used to express the data in such a way as to highlight their similarities and differences.

Background Statistics

→ Mean

$$\bar{X} = \frac{\sum_{i=1}^{n} X_i}{n}$$

→ Standard Deviation

$$s = \sqrt{\frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{(n-1)}}$$

→ Variance

$$s^2 = \frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{(n-1)}$$

Covariance

→ Standard deviation and variance can only operate one dimensional data.
→ Covariance is a measure of how much two dimensions vary from the mean with respect to each other.

$$var(X) = \frac{\sum_{i=1}^{n}(X_i - \bar{X})(X_i - \bar{X})}{(n-1)}$$

$$cov(X,Y) = \frac{\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})}{(n-1)}$$

Covariance: An Example

	Hours(H)	*Mark(M)*
Data	9	39
	15	56
	25	93
	14	61
	10	50
	18	75
	0	32
	16	85
	5	42
	19	70
	16	66
	20	80
Totals	167	749
Averages	13.92	62.42

H	M	$(H_i - \bar{H})$	$(M_i - \bar{M})$	$(H_i - \bar{H})(M_i - \bar{M})$
9	39	-4.92	-23.42	115.23
15	56	1.08	-6.42	-6.93
25	93	11.08	30.58	338.83
14	61	0.08	-1.42	-0.11
10	50	-3.92	-12.42	48.69
18	75	4.08	12.58	51.33
0	32	-13.92	-30.42	423.45
16	85	2.08	22.58	46.97
5	42	-8.92	-20.42	182.15
19	70	5.08	7.58	38.51
16	66	2.08	3.58	7.45
20	80	6.08	17.58	106.89
Total				1149.89
Average				104.54

→ **Positive value** indicates both dimensions increase together.

→ **Negative value** indicates as one increases, the other decreases.

→ Zero indicates dimensions are independent.

Covariance matrix

$$C = \begin{pmatrix} cov(x,x) & cov(x,y) & cov(x,z) \\ cov(y,x) & cov(y,y) & cov(y,z) \\ cov(z,x) & cov(z,y) & cov(z,z) \end{pmatrix}$$

Eigenvector and Eigenvalue

→ An eigenvector or characteristic vector of a square matrix is a vector that does not change its direction under the associated linear transformation.

→ If v is a vector that is not zero, then it is an eigenvector of a square matrix A if A**v** is a scalar multiple of v. This condition could be written as the equation:

$$A\mathbf{v} = \lambda\mathbf{v} \text{ where } \lambda \text{ is the eigenvalue}$$

An Example

→ Find the eigenvector and eigenvalue of the following matrix.

$$\begin{pmatrix} 2 & 3 \\ 2 & 1 \end{pmatrix}$$

Steps of PCA

→ Step 1: Get some data

→ Step 2: Subtract the mean

→ Step 3: Calculate the covariance matrix

→ Step 4: Calculate eigenvectors and eigenvalues of covariance matrix

→ Step 5: Choosing components and forming a feature vector

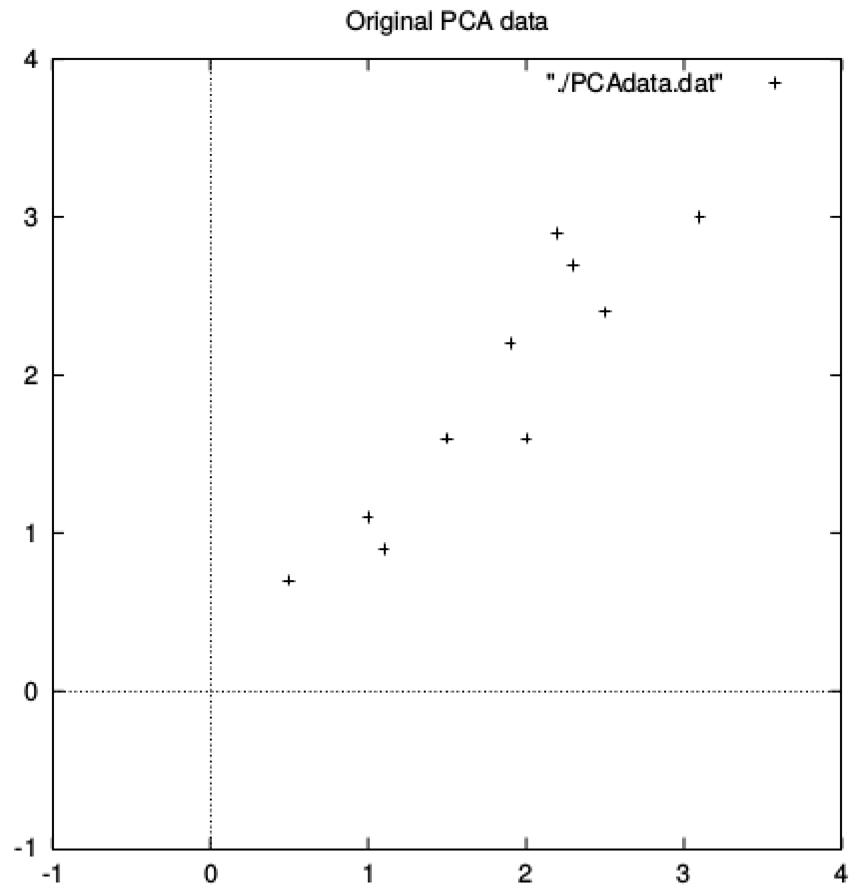
Dataset and subtracting the mean

Data =

x	y
2.5	2.4
0.5	0.7
2.2	2.9
1.9	2.2
3.1	3.0
2.3	2.7
2	1.6
1	1.1
1.5	1.6
1.1	0.9

DataAdjust =

x	y
.69	.49
-1.31	-1.21
.39	.99
.09	.29
1.29	1.09
.49	.79
.19	-.31
-.81	-.81
-.31	-.31
-.71	-1.01



Calculating the Covariance matrix

$$cov = \begin{pmatrix} .616555556 & .615444444 \\ .615444444 & .716555556 \end{pmatrix}$$

$$cov(X,Y) = \frac{\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})}{(n-1)}$$

Eigenvalues and Eigenvectors

$$eigenvalues = \begin{pmatrix} .0490833989 \\ 1.28402771 \end{pmatrix}$$

$$eigenvectors = \begin{pmatrix} -.735178656 & -.677873399 \\ .677873399 & -.735178656 \end{pmatrix}$$

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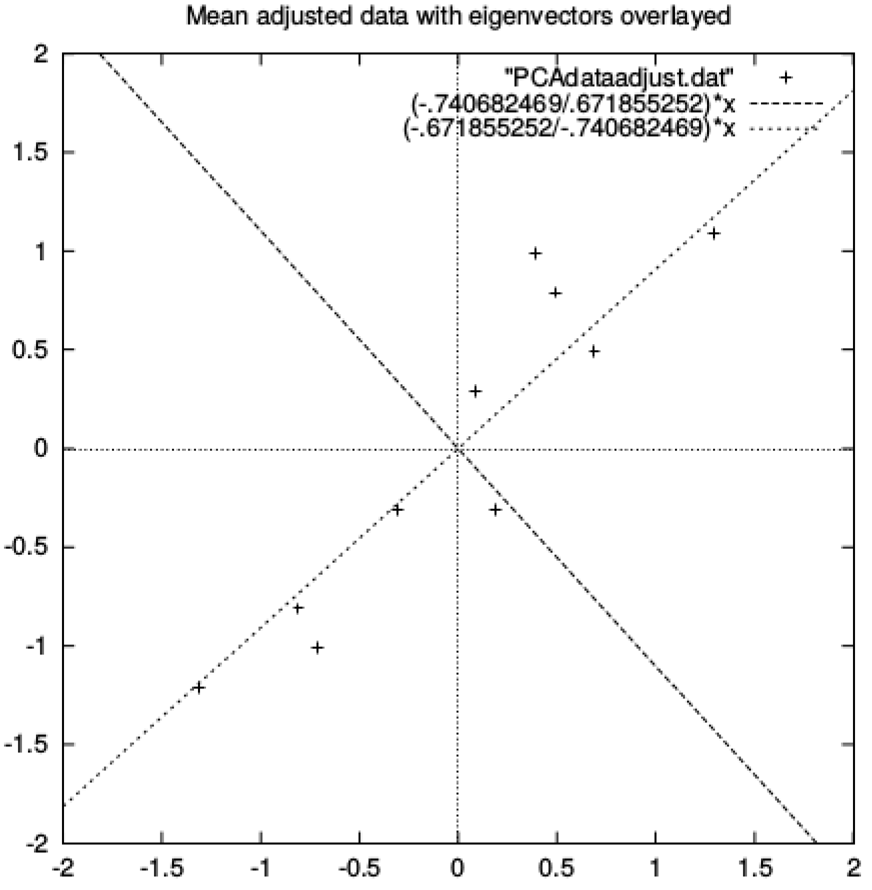

Forming a feature vector

→ The eigenvector with the highest eigenvalue is the principle component of the data set.

→ Sort the eigenvalues in descending order.

→ From n eigenvalues, take top p eigenvalues.

→ Dimensions will be changed from n to p.

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$$eigenvectors = \begin{pmatrix} -.735178656 & -.677873399 \\ .677873399 & -.735178656 \end{pmatrix}$$

$$FeatureVector = (eig_1\ eig_2\ eig_3\ ....\ eig_n)$$

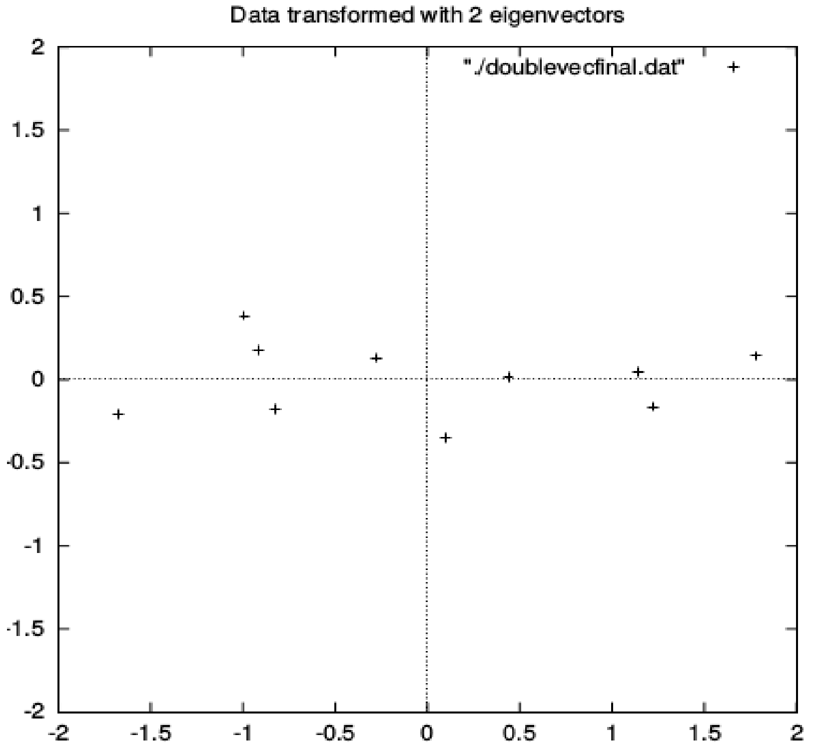
Deriving new dataset

→Take the transpose of the vector and multiply it on the left of the original data set, transposed.

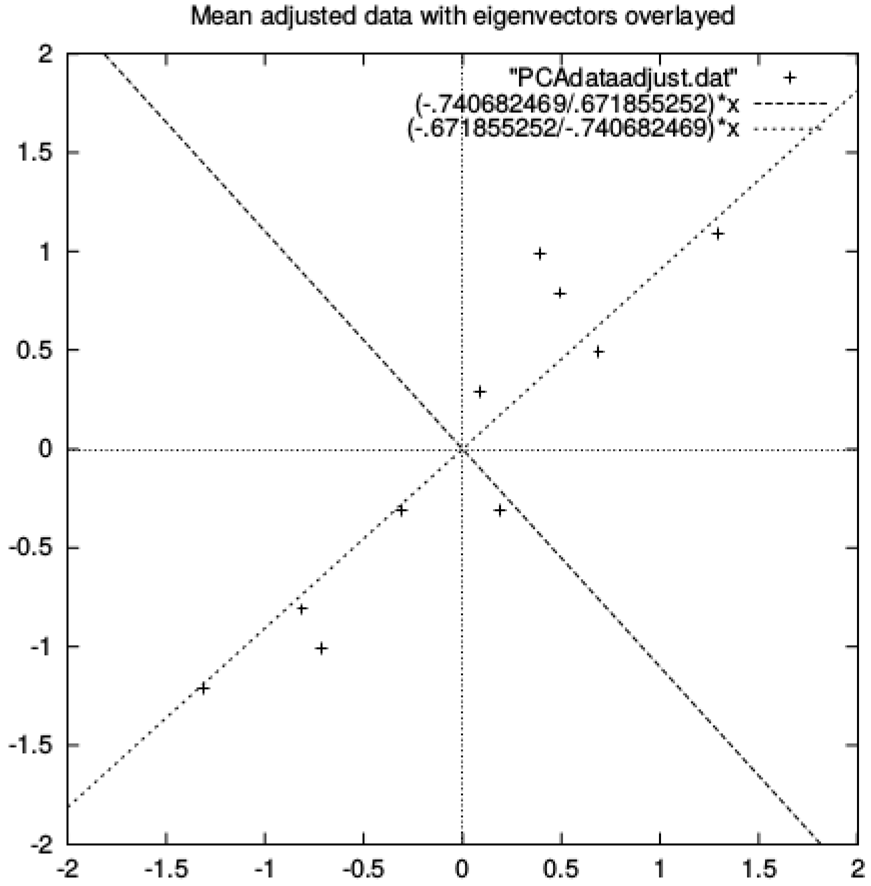
$$FinalData = RowFeatureVector \times RowDataAdjust,$$

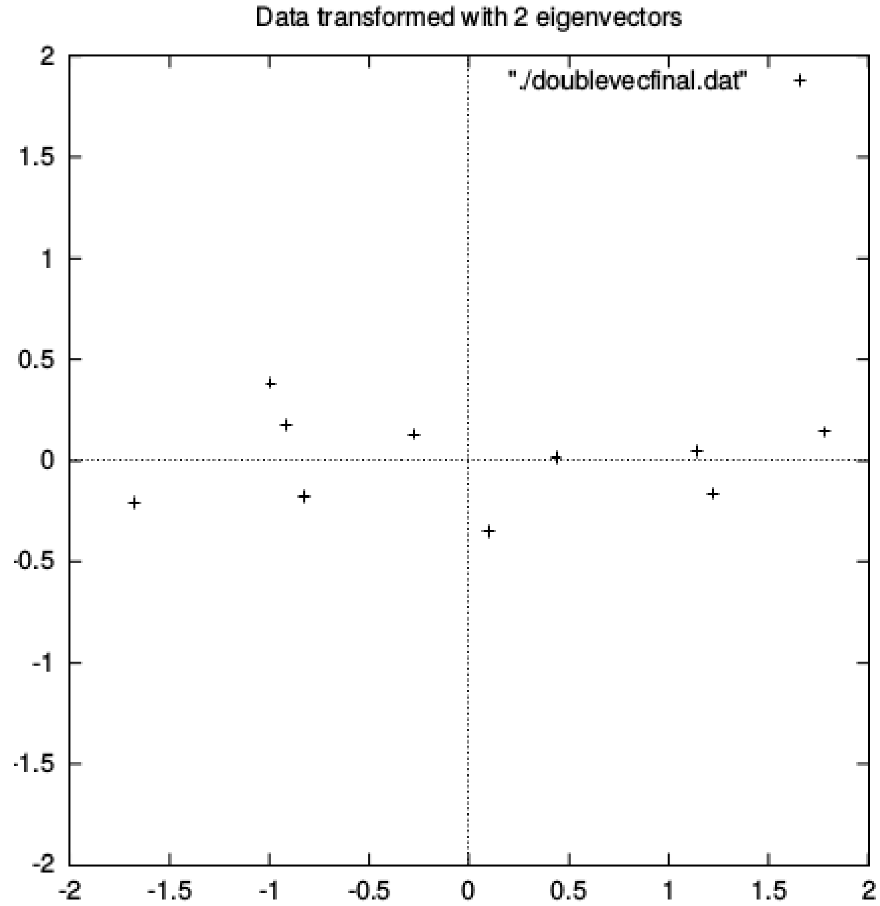
Transformed Data=

x	y
-.827970186	-.175115307
1.77758033	.142857227
-.992197494	.384374989
-.274210416	.130417207
-1.67580142	-.209498461
-.912949103	.175282444
.0991094375	-.349824698
1.14457216	.0464172582
.438046137	.0177646297
1.22382056	-.162675287



Old and new dataset





Getting Original Data back

→Getting back exact original data is only possible if we take all eigenvectors during transformation. Otherwise, we lose some information.

$$FinalData = RowFeatureVector \times RowDataAdjust,$$

$$RowDataAdjust = RowFeatureVector^{-1} \times FinalData$$

$$RowDataAdjust = RowFeatureVector^{T} \times FinalData$$

$$RowOriginalData = \left(RowFeatureVector^{T} \times FinalData\right) + OriginalMean$$